Activity characterisation of Lecitase Ultra (continuous pNPB assay)

What you're going to get / What you expect

A characterised Lecitase Ultra stock (protein concentration and specific activity) to serve as the baseline against which every immobilisation yield is referenced. The continuous assay follows p-nitrophenyl butyrate (pNPB) hydrolysis at 348 nm, a wavelength at which pNP can be read in real time (unlike the 405 nm end-point assay, which cannot be followed continuously at acidic pH).

Materials list (per measurement)

- Lecitase Ultra commercial stock.
- p-nitrophenyl butyrate (pNPB).
- Sodium phosphate buffer 25 mM, pH 7.0.
- Spectrophotometer with magnetic stirring; 1 cm cuvette.
- Bradford reagent and BSA standard (for protein quantification).

Steps

1. Add 2.5 mL of 25 mM sodium phosphate buffer pH 7.0 and 50 uL of pNPB to the cuvette under magnetic stirring.
2. Start the reaction by adding 50 uL of enzyme solution.
3. Follow pNP release continuously at 348 nm, recording the slope (change in absorbance per minute).
4. Convert the slope to activity with the equation below.
5. Quantify protein by Bradford (serial dilutions to reach the linear range) and express the specific activity as U/mg.

Activity from the continuous assay (Lambert-Beer):

$$U = (dA348 / min) / (e \cdot l)$$

$$e = 0.00515\ uM^{-1}\ cm^{-1}\ ;\ l = 1\ cm\ ;\ 1\ U = 1\ umol\ pNP\ min^{-1}$$

$$\text{Specific activity (U/mg)} = U / (c \cdot 0.05\ mL) \quad (c = mg/mL)$$

Errores comunes / troubleshooting

- Read at 348 nm, not 405 nm: at this wavelength pNP can be followed continuously, which an immobilisation time course requires.
- Very concentrated stock: dilute serially for Bradford so the reading falls in the linear range.